Experiment 5

Name: Subhradip Majumder
UID: 23BAI70059
Branch: BECSE-AIML
Section/Group: 23AIT_KRG-1

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1. **AIM:** To verify the correctness and reliability of the EcoTrack React application by writing automated tests using Jest and React Testing Library, and by analyzing application behavior using debugging tools.
2. **Objective:**
   After completing this experiment, the student will be able to:
   1. Understand the purpose of automated testing in frontend applications
   2. Write unit tests for JavaScript utility functions using Jest
   3. Use different Jest matchers to validate expected outputs and behaviors
   4. Test React components using React Testing Library
   5. Verify UI rendering by querying elements from the DOM
   6. Implement asynchronous testing using findBy and waitFor methods
   7. Apply mocking to simulate API or external data responses in tests
   8. Perform snapshot testing to detect unintended UI changes
   9. Debug failing tests and application logic using browser Developer Tools and breakpoints
   10. Analyze application behavior and errors systematically rather than manual checking

3. **Implementation/Code:**
   a. **App.test.js**

```
import { render, screen } from '@testing-library/react';
import App from './../App';
import * as productAPI from './../productAPI';

jest.mock('./../productAPI');

test("test the API function with mock", async ()=> {
  productAPI.API.mockResolvedValue({
   "name": "Subh",
   "age": 22
  });
  render(<App />);
  const nameElement = await screen.findByText(/Name:
Subh/i);
  const ageElement = await screen.findByText(/Age: 22/i);
  expect(nameElement).toBeInTheDocument();
  expect(ageElement).toBeInTheDocument();
})
```

b. [Snap.test.js]

```
import { render} from '@testing-library/react';
import Component from "./../component";


test("Snap testing", ()=>{
   const data = render(<Component data = "test"/>);
   expect(data).toMatchSnapshot();

})
```

c. [testing.test.js]

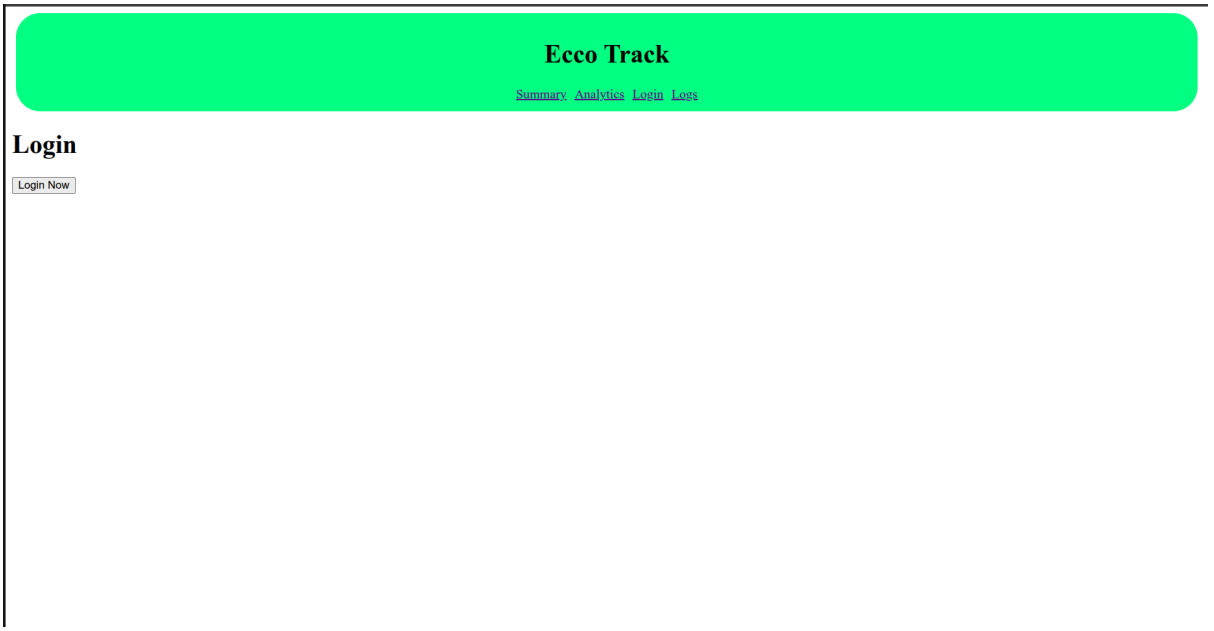
```
import { userNameCheck } from "../testing";
// Unit testing
test('Testing username', () => {
   expect(userNameCheck("")).toBe("guest");
   expect(userNameCheck("subh")).toBe("subh");
})
```

4. Output:

```
PASS  src/tests/testing.test.js
PASS  src/tests/snapApp.test.js
PASS  src/tests/App.test.js

Test Suites: 3 passed, 3 total
Tests:       3 passed, 3 total
Snapshots:   1 passed, 1 total
Time:        1.553 s
Ran all test suites.

Watch Usage: Press w to show more.
```



Ecco Track

Summary Analytics Login Logs

Dashboard_layout

DashboardSummary

Ecco Track

Summary Analytics Login Logs

Dashboard_layout

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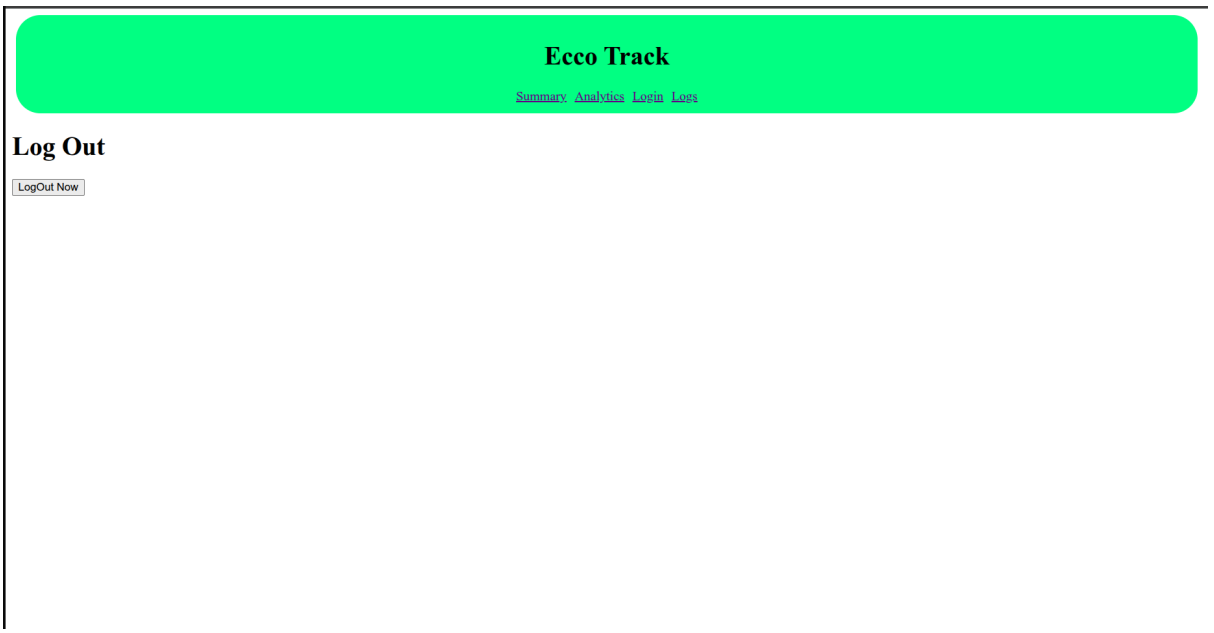

5. Learning Outcome:

a. I learnt how to make redux react.
b. How the react redux and redux slices actually work.
c. I learnt how to use the outlet tags along with nested routes.